5 Whys Root Cause Analysis

A quick way to get from a problem to its root cause by asking why until you reach something you can fix. Ideal for problems with one clear chain of cause and effect.

REFERENCE

PART NUMBER OR PROCESS

DATE

LED BY

CUSTOMER (IF ANY)

LINKED 8D OR NCR

How to use this template

1. Type straight into the boxes on screen, or print it and write by hand.
2. Do it with the people who know the process, at the place where the problem happened if you can.
3. Write facts, not opinions or blame. Every answer should be something you could check.
4. For complex problems with many possible causes, start with a fishbone diagram, then use 5 Whys on the likeliest causes.

1 State the problem

One sentence, in facts

Write what is wrong, where and how often, as a fact you could check.

- Good: "12 of 400 assemblies failed functional test on lot 4417, all at connector J3."
- Weak: "Quality is poor on the new product."

2 Ask why, five times

Each answer becomes the next question

- Start from the problem and ask why it happened. Write the answer as a fact.
- Ask why again about that answer. Keep going until you reach a cause you can act on, usually around five whys.
- Stay with the process. Stop if the answer becomes a person ("the operator forgot"): ask why the process allowed it.

If two answers are possible at any step, follow each one as a separate chain.

WHY 1? BECAUSE…

WHY 2? BECAUSE…

WHY 3? BECAUSE…

WHY 4? BECAUSE…

WHY 5? BECAUSE…

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3 Check the chain backwards

Read it from the bottom up

Read from the last why back to the problem, saying "and therefore" between each line. If any link doesn't follow, the chain is broken: go back and check that step.

Then prove the root cause with data or a trial, where you can.

HOW THE ROOT CAUSE WAS CHECKED

4 Fix it

One action for the root cause, one for why it wasn't caught

- Choose an action that removes the root cause, not one that only deals with the symptom.
- Ask one more question: why didn't our checks catch it? Add an action for that too.
- Give every action an owner and a date.

ACTION	OWNER	DUE	DONE

Going round in circles on a recurring problem?

Grindstone leads root cause investigations alongside your team, on site or remotely, at a fixed price agreed before we start.

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